

The grazing-management influence diagram

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1 Overview

SPEC section 46 extends the reference habitat network with a grazing-management decision, an uncertain implementation and two utilities:

```
Climate -> ClimateForecast           Climate -> SoilMoisture
CurrentVegetation -----> Vegetation -> HabitatQuality -> Occupancy
-> Biodiversity
GrazingManagement (decision) -> GrazingPressure -> Vegetation
information: ClimateForecast, CurrentVegetation -> GrazingManagement
utilities:  ConservationBenefit(Biodiversity), ManagementCost(GrazingManagement)
```

The decision acts on the vegetation only through GrazingPressure, whose kernel encodes imperfect implementation (SPEC section 42): excluding grazing yields low pressure with probability 0.95, maintaining it with probability 0.15. This vignette runs the five analyses required by SPEC section 46 on the zoo's reference model, `reference_grazing_id()`, which is `InfluenceDiagrams.reference_grazing_model()`.

2 Setup

```
using EcologicalBayesianNetworks
using InfluenceDiagrams
g = reference_grazing_id()
```

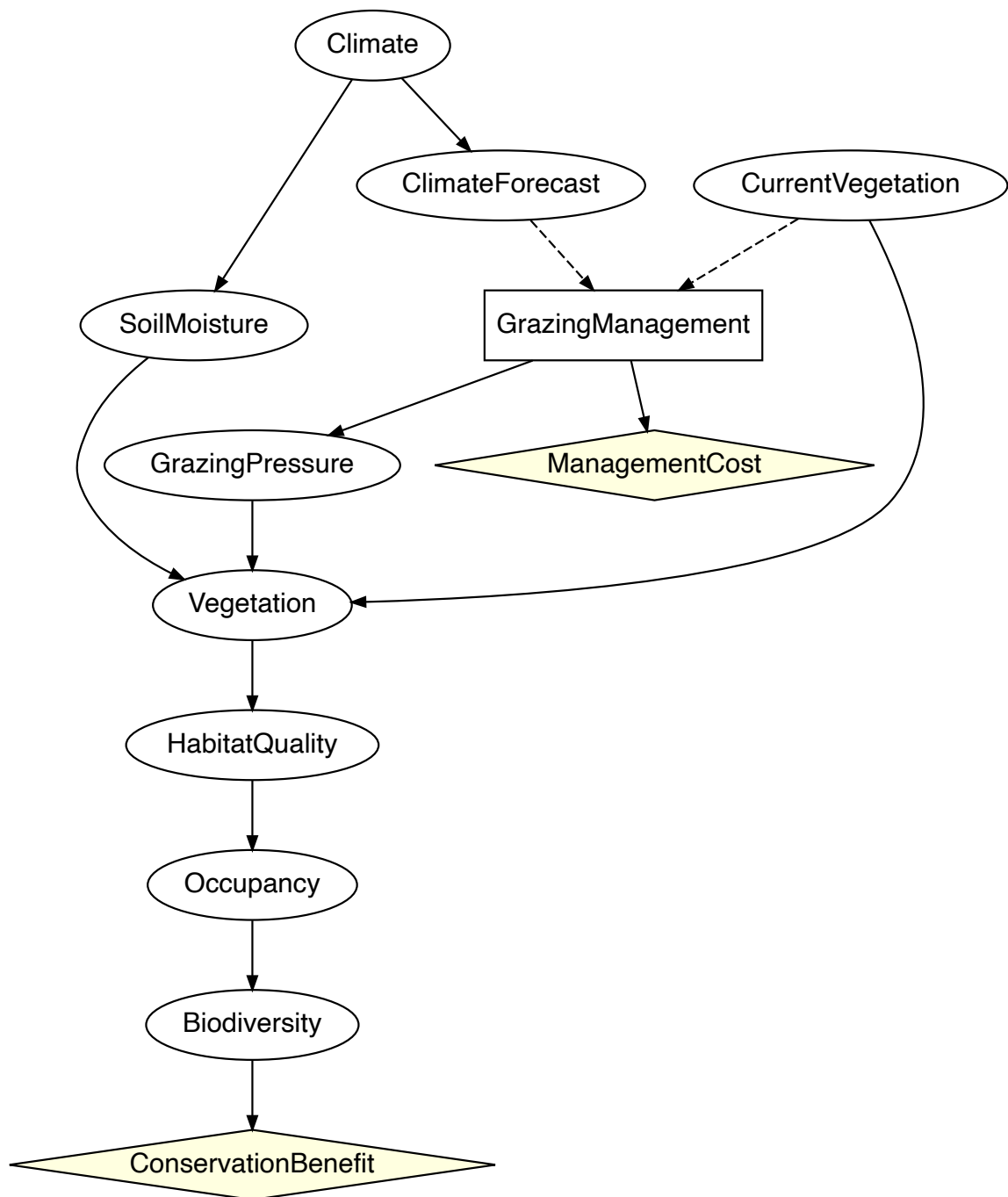
```
InfluenceDiagramModel(10 variables, 1 decision, 2 utilities, 9 kernels, 2 bound)
```

The same model comes back from `load_model`, and from the Netica fixture of `BayesianNetworkFormats.jl` through `read_influence_diagram`:

```
load_model("reference_grazing_id") ≈ g,
read_influence_diagram(fixture_path("dne/grazing-reference_id.dne")) ≈ g
```

```
(true, true)
```

```
to_graphviz(g; states = false)
```



Decisions are boxes and utilities diamonds; dashed arcs are information arcs, which are not causal (SPEC section 34).

```
information_names(syntax(g), :GrazingManagement), utility_names(syntax(g))
```

```
([:ClimateForecast, :CurrentVegetation], [:ConservationBenefit, :ManagementCost])
```

```
model_summary("reference_grazing_id")
```

```
ModelSummary reference_grazing_id (Reference grazing-management influence diagram
(SPEC section 46))
  format / licence      julia / MIT (builtin)
  nodes                 12 (9 chance, 1 decision, 2 utility)
  arcs                  13
  largest state space    3
  largest in-degree      3
  load_model             InfluenceDiagramModel
```

3 Analysis 1: expected utility of each fixed action

`expected_utility(g, :D ⇒ :a)` fixes the decision (a constant policy) and evaluates the additive utility, conservation benefit minus management cost, by brute force over the joint law.

```
actions = (:exclude, :reduce, :maintain)
fixed = [a ⇒ expected_utility(g, :GrazingManagement ⇒ a) for a in actions]
```

```
3-element Vector{Pair{Symbol, Float64}}:
 :exclude => 2.422243462499958
 :reduce  => 24.84182619999995
 :maintain => 36.524146862500004
```

The benefit is 100 times the probability of high biodiversity; the cost is 40, 15 or 0.

```
[a ⇒ 100 * marginal(instantiate(fix_decision(g, :GrazingManagement ⇒ a)),
:Biodiversity).table[2]
for a in actions]
```

```
3-element Vector{Pair{Symbol, Float64}}:
 :exclude => 42.422243462499985
 :reduce  => 39.8418262
 :maintain => 36.5241468625
```

4 Analysis 2: the optimal conditional policy

`optimize` runs decision variable elimination by default and returns the maximal expected utility with an optimal strategy.

```
sol = optimize(g)
sol.expected_utility
```

```
36.524146862500004
```

The policy table is indexed by the information set in order: rows are the climate forecast (dry, normal, wet), columns the current vegetation (sparse, moderate, dense). With the reference numbers the management cost (40 for exclusion) dominates a conservation benefit of at most 100, so the optimal policy maintains grazing whatever the information, and the maximal expected utility equals the best fixed action:

```
policy_table(sol.strategy[:GrazingManagement])
```

```
3×3 Matrix{Symbol}:
:maintain :maintain :maintain
:maintain :maintain :maintain
:maintain :maintain :maintain
```

The optimum is confirmed by exhaustive search over all 3^9 deterministic strategies:

```
optimize(g, ExhaustivePolicySearch()).expected_utility, maximum(last.(fixed))
```

```
(36.52414686249998, 36.524146862500004)
```

A decision problem is only interesting when the information can change the decision. Valuing high biodiversity ten times more (a reserve, say) is a one-line change of the utility bound to the ConservationBenefit node; the syntax, the kernels and the information structure are untouched. The optimal policy now reacts to both the forecast and the survey:

```
h = bind_utility(g, :ConservationBenefit => [0.0, 1000.0])
solh = optimize(h)
solh.expected_utility, policy_table(solh.strategy[:GrazingManagement])
```

```
(384.64721647500005, [:reduce :exclude :exclude; :exclude :exclude :reduce; :exclude
:exclude :reduce])
```

```
[a => expected_utility(h, :GrazingManagement => a) for a in actions]
```

```
3-element Vector{Pair{Symbol, Float64}}:
:exclude => 384.22243462499995
:reduce => 383.41826199999963
:maintain => 365.24146862500003
```

Reacting is not the same as being worth reacting to, and it is worth saying plainly how little is at stake here. The conditional policy earns about 0.4 more than the best fixed action on an objective of about 385, roughly a tenth of a percent:

```
best_fixed = maximum(expected_utility(h, :GrazingManagement => a) for a in actions)
(optimum = solh.expected_utility, best_fixed = best_fixed,
 gain = solh.expected_utility - best_fixed,
 gain_percent = 100 * (solh.expected_utility - best_fixed) / solh.expected_utility)
```

```
(optimum = 384.64721647500005, best_fixed = 384.22243462499995, gain = 0.4247818500001017, gain_percent = 0.11043414115742345)
```

The remaining analyses use this valuation, and the reader should carry that number through them: everything Analysis 4 measures is of that size.

5 Analysis 3: dropping the vegetation survey

`without_information` removes an information arc and drops the policies whose signature changed. Without the survey the policy can only react to the forecast:

```
nosurvey = without_information(h, :GrazingManagement, :CurrentVegetation)
sol3 = optimize(nosurvey)
information_names(syntax(nosurvey), :GrazingManagement), sol3.expected_utility,
policy_table(sol3.strategy[:GrazingManagement])
```

```
([:ClimateForecast], 384.22243462500006, [:exclude, :exclude, :exclude])
```

6 Analysis 4: the value of observing the current vegetation

The value of the survey is the difference between the two optima (SPEC section 35) – the expected value of information of Howard (1966), and the quantity Runge et al. (2011) argue should be computed before an adaptive-management programme commissions any monitoring:

```
expected_value_of_information(nosurvey, :CurrentVegetation, :GrazingManagement)
```

```
0.424781849999988
```

The forecast is worth less than the survey, and perfect information about everything upstream of the decision bounds what any monitoring programme could be worth:

```
(forecast = expected_value_of_information(without_information(h, :GrazingManagement,
:ClimateForecast),
:ClimateForecast, :GrazingManagement),
soil_moisture = expected_value_of_information(h, :SoilMoisture,
:GrazingManagement),
perfect = expected_value_of_perfect_information(h, :GrazingManagement))
```

```
(forecast = 0.101836350000004188, soil_moisture = 0.6800549000000016, perfect = 0.6800549000000016)
```

These are the honest numbers and they are almost nothing: about 0.4 for the survey, 0.1 for the forecast and 0.7 for perfect information, against an objective near 385. A monitoring programme costing more than a rounding error on this objective would not pay for itself. The rescaled valuation makes the policy react to the information; it does not make the information worth having.

6.1 Why no rescaling of the benefit alone can fix that

The reason is structural, not a matter of choosing a bigger number. The additive utility is $B * P(\text{Biodiversity} = \text{high} \mid \text{action}) + \text{cost}(\text{action})$, so the optimal action in information state i is whichever maximises $B * P_a(\text{high} \mid i) + \text{cost}(a)$. Information is worth something only when that argmax changes with i , which needs the *difference* between actions, not their level, to depend on the information. Here it barely does:

```
fcs, vgs = (:dry, :normal, :wet), (:sparse, :moderate, :dense)
p_high(a, f, v) = probability(marginal(observe(instantiate(fix_decision(g,
:GrazingManagement => a)),
                                [:ClimateForecast => f,
:CurrentVegetation => v]),
                             :Biodiversity), :high)
gaps = [round(p_high(:exclude, f, v) - p_high(:maintain, f, v); digits = 4)
        for f in fcs, v in vgs]
```

```
3×3 Matrix{Float64}:
 0.056  0.0632  0.0585
 0.0594 0.0626  0.0548
 0.0613 0.0608  0.0496
```

Excluding grazing raises the chance of high biodiversity by between about 0.050 and 0.063 whatever the forecast and whatever the survey says: a spread of a quarter around a mean of about 0.059. Multiplying the benefit by B multiplies both the spread and the objective, so the ratio between them is fixed. Sweeping B from 100 to 1600 never lifts the value of perfect information above about 1 on an objective that grows past 600.

6.2 A third valuation, where the information is decisive

What *can* be tuned is the cost vector, because information is worth most when two actions are nearly tied. On average, excluding grazing beats reducing it by about 25.8 in benefit at $B = 1000$; pricing exclusion at 51 against reduction at 25 makes the two almost exactly break even, so which of them wins is decided state by state rather than on average:

```
k = bind_utility(g, [:ConservationBenefit => [0.0, 1000.0],
                   :ManagementCost => [-51.0, -25.0, 0.0])
fixed_k = [a => expected_utility(k, :GrazingManagement => a) for a in actions]
```

```
3-element Vector{Pair{Symbol, Float64}}:
 :exclude => 373.22243462499966
 :reduce  => 373.4182619999997
 :maintain => 365.2414686250003
```

```
solk = optimize(k)
solk.expected_utility, policy_table(solk.strategy[:GrazingManagement])
```

```
(373.9873744375, [:reduce :exclude :reduce; :reduce :exclude :reduce; :exclude
:exclude :reduce])
```

The policy now genuinely splits: exclusion on a moderate vegetation survey and on a wet forecast with sparse vegetation, reduction elsewhere. The value of information rises, and - more to the point - so does its share of what the decision is worth at all. Comparing the value of information with the whole objective was always the wrong denominator: most of that objective is earned no matter what the manager does. The quantity a monitoring budget competes against is the spread between the best and the worst action.

```
best_k, worst_k = maximum(last.(fixed_k)), minimum(last.(fixed_k))
(gain = solk.expected_utility - best_k,
 survey = expected_value_of_information(without_information(k, :GrazingManagement,
:CurrentVegetation),
                                     :CurrentVegetation, :GrazingManagement),
 forecast = expected_value_of_information(without_information(k, :GrazingManagement,
:ClimateForecast),
                                     :ClimateForecast, :GrazingManagement),
 perfect = expected_value_of_perfect_information(k, :GrazingManagement),
 decision_range = best_k - worst_k)
```

```
(gain = 0.5691124375003369, survey = 0.5400709624999536, forecast =
0.0607149374999949, perfect = 1.0309598124999866, decision_range =
8.176793374999363)
```

Perfect information is worth about 1.0 where the decision itself is worth about 8.2, so around an eighth of the value at stake. This is an upper bound on a monitoring programme's benefit, not a commissioning recommendation: a real signal may be much less valuable. Against the full objective of about 374 the same number is 0.3%, which is why the denominator and the actual signal both have to be stated.

6.3 Price the imperfect signal, not the perfect-information bound

Use the uninformed umbrella model to keep the comparison transparent. A forecast has a noisy observation mechanism; perfect weather knowledge is a different signal. Both gains below are measured against the same no-information baseline:

```
uninformed = without_information(umbrella_model(), :Umbrella, :Forecast)
forecast_value = expected_value_of_information(uninformed, :Forecast, :Umbrella)
perfect_value = expected_value_of_perfect_information(uninformed, :Umbrella)
forecast_cost = 10.0
(forecast_value = forecast_value, perfect_value = perfect_value,
 cost = forecast_cost, net_value = forecast_value - forecast_cost,
 commission = forecast_value > forecast_cost)
```

```
(forecast_value = 7.000000000000014, perfect_value = 21.0, cost = 10.0, net_value =
-2.999999999999986, commission = false)
```

The forecast is worth 7, perfect information 21, and this forecast costs 10. It should not be commissioned even though its cost is below EVPI. In a substantive application the signal's accuracy, timing and cost must be specified before its expected information value can be compared with cost.

The remaining analyses stay with `h`, the plain tenfold valuation, so that they compare with Analyses 2 and 3.

7 Analysis 5: direct intervention versus deciding under uncertain implementation

A hard intervention `do(GrazingPressure = low)` replaces the mechanism of the grazing pressure by a point mass: the pressure is low whatever the decision, and the decision only carries its cost. A management decision followed by uncertain implementation is the diagram as read: choosing `exclude` makes low pressure likely but not certain, and the cost is paid regardless.

```
direct = do_intervention(h, :GrazingPressure => :low)
intervened_variables(direct), history(direct)[1].kind
```

```
([:GrazingPressure], :hard)
```

Under the intervention the management decision no longer affects the ecology, so the optimal policy is the cheapest action, `maintain`:

```
sol_direct = optimize(direct)
sol_direct.expected_utility,
unique(policy_table(sol_direct.strategy[:GrazingManagement]))
```

```
(427.908745, [:maintain])
```

The conservation benefit of guaranteed low pressure compared with excluding grazing (the closest a manager can get to `do(GrazingPressure = low)`) and with maintaining it measures what perfect implementation would be worth:

```
benefit(m, a) = 1000 * marginal(instantiate(fix_decision(m, :GrazingManagement =>
a)), :Biodiversity).table[2]
(do_low = benefit(direct, :maintain), exclude = benefit(h, :exclude), maintain =
benefit(h, :maintain))
```

```
(do_low = 427.908744999999984, exclude = 424.2224346249999, maintain = 365.241468625)
```

`do_intervention` on the action variable itself is different again: it removes the decision and installs a constant mechanism, the physical override of SPEC section 41, distinct from `fix_decision`, which keeps the decision and its information set and only changes the strategy.

```
override = do_intervention(h, :GrazingManagement => :exclude)
nparts(syntax(override), :Decision),
expected_utility(instantiate(override)) ≈ expected_utility(h, :GrazingManagement =>
:exclude)
```

(0, true)

8 Summary

Adding a decision, an information set and two utilities to the reference habitat network turns it into an influence diagram that the same machinery solves, and the five SPEC section 46 analyses run without leaving the zoo. The honest finding is that with the reference numbers the optimal policy is constant and the value of the vegetation survey is a rounding error on the objective: information is worth nothing when it cannot change the decision, which is exactly what a value-of-information analysis is for (Runge et al. 2011). The next vignette builds the implementation chain of SPEC section 42 explicitly, as an open network glued onto the habitat network.

9 References

- Howard, Ronald A. 1966. “Information Value Theory.” *IEEE Transactions on Systems Science and Cybernetics* 2 (1): 22–26. <https://doi.org/10.1109/TSSC.1966.300074>.
- Runge, Michael C., Sarah J. Converse, and James E. Lyons. 2011. “Which Uncertainty? Using Expert Elicitation and Expected Value of Information to Design an Adaptive Program.” *Biological Conservation* 144 (4): 1214–23. <https://doi.org/10.1016/j.biocon.2010.12.020>.